

Market Data Types and Data Structures

Recognize market-data objects and preserve identity, timing, and durable representations.

Library of the Quant

Educational reference manual

Manual Profile

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Educational Use Only

This manual is educational material for quant research study. It is not financial advice, not a trading recommendation, and not a guarantee of correctness or live trading usefulness.

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How to use this manual

BEGINNER

1. IDENTIFY THE OBJECT

Start by naming whether the record is a trade, quote, book message, bar, action, reference fact, or derived estimate. Each object has different identity and correction rules.

2. DECLARE THE GRAIN

Say exactly what one row represents and which fields make it unique. Check the cardinality of every join before trusting totals, counts, or cross-sectional samples.

3. SEPARATE THE CLOCKS

Trace event, arrival, processing, publication, decision, and fill times. Historical eligibility follows the system's knowledge clock, not a convenient final timestamp.

4. PRESERVE RAW AND DERIVED LAYERS

Keep immutable source events, normalized contracts, and versioned derived datasets distinct. Cleaning should add flags and views rather than erase evidence.

5. CHOOSE ACCESS BY WORKLOAD

Match long, wide, row, column, stream, state, and snapshot structures to actual queries. Benchmark scans, joins, replay, and recovery using representative skew.

6. TEST ECONOMIC MEANING

Schema, null, and key tests are necessary but insufficient. Reconcile units, calendars, venue coverage, corporate actions, freshness, and point-in-time behavior.

DECISION STANDARD

A dataset is research-ready only when identity, time, grain, units, quality, lineage, snapshot, ownership, and failure behavior are reproducible.

Notation and record contract

BEGINNER

SYMBOL	OBJECT	MEANING
t_e	event time	When the market or source says the event occurred
t_a	arrival time	When the collector received the payload
t_p	publish time	When normalized data became available
id	instrument identity	Stable internal security or contract identifier
seq	sequence	Source ordering within a session or channel
p, q	price and quantity	Values with currency, unit, and multiplier
[b0,b1)	interval	Left-closed, right-open aggregation window
age	freshness	Decision time minus last observed time
v	contract version	Schema and semantic interpretation ID
s	snapshot	Immutable logical dataset state

RECORD IDENTITY

A robust event key combines source identity, instrument, event type, source sequence, and time. A database-generated row number cannot prove two deliveries represent the same economic event.

MINIMUM DATASET METADATA

Store grain, key, schema, units, calendar, clocks, correction policy, source cutoff, snapshot, row counts, quality report, owner, retention, and downstream contract.

Market data to research dataset

BEGINNER

1

Acquire

Raw payload, license, source clocks, and checksums

2

Resolve identity

Instrument, venue, contract, and validity interval

3

Normalize

Typed fields, units, condition codes, and event keys

4

Order and reconcile

Sequence gaps, duplicates, corrections, and state

5

Validate

Schema, economics, freshness, coverage, and alternate source

6

Derive

Books, bars, adjustments, universes, and point-in-time facts

7

Snapshot

Atomic manifest, lineage, schema, and test report

8

Serve

As-of API, analytical files, stream, and current state

9

Monitor

Drift, SLA, reconciliation, consumer impact, and fallback

HARD GATE

No dataset reaches a model or capital decision until its point-in-time eligibility, quality, units, and snapshot lineage can be replayed.

The market-data contract

BEGINNER

ONE-SENTENCE MEANING

A market-data contract defines what one record means, when it became knowable, which instrument it describes, and how its fields and units must be interpreted.

MEMORY JOGGER

A numeric table is not yet data you can trust. Every row needs an identity, clock, grain, schema, and availability rule.

WHY IT MATTERS IN QUANT RESEARCH

Research results can change when a vendor changes a price field, timestamp convention, correction policy, or symbology map. Explicit contracts make those changes visible and testable.

FORMULA INTUITION

A useful record key is often (source, instrument_id, event_time, sequence, event_type). Uniqueness and ordering expectations belong beside the field definitions.

SIMPLE MARKET EXAMPLE

Two vendors both publish `close`, but one uses the last eligible trade and the other an official auction price. A close-to-close signal will not reproduce until the distinction is recorded.

PYTHON / SYSTEM IMPLEMENTATION

Represent contracts as versioned schemas with type, nullability, unit, timezone, allowed values, key, lateness, correction, and owner metadata. Validate records at ingestion and again at consumption boundaries.

CONFUSIONS TO AVOID

Column names are not definitions. Treating every timestamp as exchange time or every price as executable creates silent, systematic error rather than an obvious exception.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The ingestion registry binds feed versions to normalized tables; downstream datasets carry the contract ID and reject incompatible versions before research or trading.

RELATED CONCEPTS AND QUICK RECALL

Related: schema, grain, event identity, data lineage. Recall task: Write the minimum contract for one quote update and name the fields that prove its identity and availability.

Event time, arrival time, and processing time

BEGINNER

ONE-SENTENCE MEANING

Market systems need separate clocks for when an event occurred, when it reached a collector, and when a pipeline processed or published it.

MEMORY JOGGER

One event can have several legitimate timestamps. The correct clock depends on whether you are reconstructing the market, measuring latency, or proving what a strategy knew.

WHY IT MATTERS IN QUANT RESEARCH

Using exchange event time alone can make a historical feature appear available before the network delivered it. Arrival and publish clocks support causal replay and realistic execution assumptions.

FORMULA INTUITION

Latency components can be written $L_{\text{network}} = t_{\text{arrival}} - t_{\text{event}}$ and $L_{\text{pipeline}} = t_{\text{publish}} - t_{\text{arrival}}$; total observable delay is $t_{\text{publish}} - t_{\text{event}}$.

SIMPLE MARKET EXAMPLE

A quote stamped 10:00:00.002 at the venue arrives at 10:00:00.006 and enters the research bus at 10:00:00.011. A decision at 10:00:00.008 cannot use the published normalized record.

PYTHON / SYSTEM IMPLEMENTATION

Store UTC nanosecond timestamps plus source timezone metadata; never overwrite one clock with another. Monitor negative or implausible latency and preserve sequence numbers for tie-breaking.

CONFUSIONS TO AVOID

Sorting by a single timestamp does not reconstruct causal order when clocks drift or several events share a timestamp. Processing time is also not a substitute for venue sequence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The replay engine chooses an explicit knowledge clock, while latency analytics compare event, arrival, normalize, feature, decision, route, and fill timestamps.

RELATED CONCEPTS AND QUICK RECALL

Related: latency, causal replay, sequence number, clock synchronization. Recall task: Which timestamp determines market order, and which determines feature eligibility for your own system?

Trade prints and sale conditions

BEGINNER

ONE-SENTENCE MEANING

A trade record reports an executed transaction, but its economic meaning depends on venue, price, size, aggressor inference, and sale-condition flags.

MEMORY JOGGER

A print says that something traded, not necessarily that the displayed market moved or that the trade was regular and immediately actionable.

WHY IT MATTERS IN QUANT RESEARCH

Trades support returns, volume, volatility, impact, order-flow, and execution research. Filtering corrections, auctions, late reports, and special conditions prevents distorted bars and labels.

FORMULA INTUITION

Dollar volume is $\sum_i p_i q_i$; VWAP is $\sum_i p_i q_i / \sum_i q_i$ over a declared eligibility window.

SIMPLE MARKET EXAMPLE

A large late-reported block can arrive during the session with an old execution time. Aggregating by arrival places it in the wrong bar; ignoring its condition can create a false price jump.

PYTHON / SYSTEM IMPLEMENTATION

Persist raw condition codes and the normalized eligibility decision. Build separate views for consolidated tape, venue-native trades, regular-session bars, auctions, and all-print audit.

CONFUSIONS TO AVOID

Last trade is not always the best estimate of current fair value. Bid-ask bounce, odd lots, off-exchange prints, corrections, and delayed reports can dominate high-frequency returns.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The canonical trade table feeds bar builders and microstructure features through policy-specific eligibility views rather than destructive cleaning of the raw event log.

RELATED CONCEPTS AND QUICK RECALL

Related: VWAP, tick rule, aggressor side, auction print. Recall task: Explain why a late-reported block needs both execution and arrival timestamps before it can enter a bar.

Quotes and the national best market

BEGINNER

ONE-SENTENCE MEANING

A quote advertises available bid and ask prices and sizes at a moment, while a consolidated best market selects the highest eligible bid and lowest eligible ask across venues.

MEMORY JOGGER

Quotes describe displayed intent, not guaranteed fills. They can update or cancel faster than a strategy can react.

WHY IT MATTERS IN QUANT RESEARCH

Quote data supports spread, midpoint, microprice, liquidity, routing, and mark calculations. Venue provenance and eligibility rules determine whether a displayed price was reachable.

FORMULA INTUITION

Midpoint $m = (\text{bid} + \text{ask}) / 2$; spread $s = \text{ask} - \text{bid}$; quoted depth near top is $\text{bid_size} + \text{ask_size}$, subject to consistent units.

SIMPLE MARKET EXAMPLE

Venue A shows 100 shares at 50.00 bid while venue B shows 200 at 50.01. The best bid is 50.01, but a 500-share sell order faces more than the single displayed top level.

PYTHON / SYSTEM IMPLEMENTATION

Normalize quote action, side, price, size, venue, participant, sequence, and condition fields. Reconstruct state from updates and snapshot it only after ordering and gap checks.

CONFUSIONS TO AVOID

A crossed or locked consolidated quote can reflect latency, different eligibility rules, or a real transient state. Blindly replacing such values with nulls removes useful quality evidence.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The quote state service publishes venue books, consolidated best quotes, quality flags, and freshness so research can choose the appropriate executable abstraction.

RELATED CONCEPTS AND QUICK RECALL

Related: NBBO, midpoint, spread, quote fade. Recall task: When is the midpoint a better mark than the last trade, and when is it still not executable?

Order-book levels: L1, L2, and L3

BEGINNER

ONE-SENTENCE MEANING

Order-book data ranges from the best bid and ask, to aggregated depth by price, to individual order messages with queue identity.

MEMORY JOGGER

More depth is not automatically more truth. Each level adds state, sequencing, and venue-specific interpretation requirements.

WHY IT MATTERS IN QUANT RESEARCH

L1 supports basic spread and mark research; L2 supports depth, imbalance, and sweep-cost estimates; L3 supports queue position and order-level dynamics where identifiers are reliable.

FORMULA INTUITION

A common imbalance is $I = (Q_{\text{bid}} - Q_{\text{ask}}) / (Q_{\text{bid}} + Q_{\text{ask}})$, computed over a declared set of levels and units.

SIMPLE MARKET EXAMPLE

Top-of-book imbalance may be strongly positive while deeper sell liquidity dominates. A market buy larger than displayed ask size walks several levels, so L1 understates expected impact.

PYTHON / SYSTEM IMPLEMENTATION

Use an append-only message table plus deterministic state builder. Verify sequence continuity, apply adds/modifies/cancels/trades, and checkpoint state for restart and replay.

CONFUSIONS TO AVOID

Depth snapshots cannot reveal event order or queue priority. L3 identifiers may reset, recycle, or represent broker aggregates, so their semantics must be documented per venue.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The execution simulator consumes a versioned book representation whose reconstruction health, depth coverage, and staleness are explicit inputs to fill logic.

RELATED CONCEPTS AND QUICK RECALL

Related: market depth, queue position, book imbalance, sweep cost. Recall task: State what additional question L3 can answer that an L2 snapshot cannot, and name its main reconstruction risk.

OHLCV bars and aggregation policy

BEGINNER

ONE-SENTENCE MEANING

An OHLCV bar summarizes eligible events over a declared interval using first, maximum, minimum, last price, and accumulated volume.

MEMORY JOGGER

A bar is a lossy compression of an event path. Its meaning comes from interval boundaries, eligible trades, price field, and empty-window policy.

WHY IT MATTERS IN QUANT RESEARCH

Bars reduce storage and simplify feature engineering, but they hide within-bar order, spreads, depth, and multiple possible paths sharing the same summary.

FORMULA INTUITION

For event set E_b : $O = \text{first}(p)$, $H = \max(p)$, $L = \min(p)$, $C = \text{last}(p)$, $V = \text{sum}(q)$. The set must be defined before the formulas are meaningful.

SIMPLE MARKET EXAMPLE

A minute with prices 100, 102, 99, 101 and volumes 10, 5, 8, 12 produces the same OHLC as several different sequences with different stop and fill outcomes.

PYTHON / SYSTEM IMPLEMENTATION

Write a deterministic bar builder with exchange calendar, boundary convention, eligibility filter, late-event policy, empty-bar rule, and reproducible revisions. Keep source-event lineage for audits.

CONFUSIONS TO AVOID

Forward-filling a close can be useful for a regular grid but creates a synthetic observation. It must not create volume, an eligible trade, or evidence of liquidity.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The normalized event store is authoritative; bars are versioned derived products keyed by policy so feature stores can reproduce the exact research grid.

RELATED CONCEPTS AND QUICK RECALL

Related: time bar, volume bar, dollar bar, resampling. Recall task: List four policy choices required to reproduce a one-minute OHLCV series.

Corporate actions and adjusted prices

BEGINNER

ONE-SENTENCE MEANING

Corporate-action data describes events such as splits, dividends, mergers, spinoffs, and symbol changes that alter the economic interpretation of price and shares.

MEMORY JOGGER

A raw price discontinuity may be an accounting event rather than a market return. Adjustment is a transformation with a timestamp and policy, not a cosmetic cleanup.

WHY IT MATTERS IN QUANT RESEARCH

Return research, universe history, shares outstanding, and factor exposures depend on consistent action treatment. Point-in-time announcement and effective dates also matter for event studies.

FORMULA INTUITION

A split-adjusted historical price often uses $P_{adj,t} = P_{raw,t} / A_t$, while total-return series also reinvest distributions under a stated convention.

SIMPLE MARKET EXAMPLE

A two-for-one split halves the raw price and doubles shares. Treating the price change as a -50% loss creates extreme returns, false volatility, and spurious stop events.

PYTHON / SYSTEM IMPLEMENTATION

Store actions as effective-dated events with vendor version, announcement time, ex-date, payment date, factors, cash terms, and affected identifiers. Generate raw, split-adjusted, and total-return views separately.

CONFUSIONS TO AVOID

Back-adjusted history can change after new actions or vendor corrections. It is unsuitable as immutable raw truth and may leak future knowledge into historical shares or membership.

WHERE THIS FITS IN THE RESEARCH SYSTEM

A corporate-action service publishes point-in-time factors and mapping events to bar, returns, universe, and fundamentals pipelines with explicit adjustment policy IDs.

RELATED CONCEPTS AND QUICK RECALL

Related: split factor, dividend, total return, adjustment chain. Recall task: Why must a research store preserve raw prices even when most models use adjusted returns?

Reference data and instrument identity

BEGINNER

ONE-SENTENCE MEANING

Reference data maps stable internal instruments to changing symbols, venues, contracts, currencies, and issuer relationships over effective time.

MEMORY JOGGER

Tickers are labels; instruments are identities. The same ticker can be reused, and one issuer can have many tradable securities.

WHY IT MATTERS IN QUANT RESEARCH

Correct joins, roll histories, delistings, corporate actions, and cross-vendor reconciliation depend on stable identity. Identity errors often look like impossible returns or missing history.

FORMULA INTUITION

A temporal mapping is valid when $\text{effective_from} \leq \text{event_time} < \text{effective_to}$; joins must use both identifier and validity interval.

SIMPLE MARKET EXAMPLE

A ticker changes after a merger while the vendor permanently backfills the new ticker. Joining only on text can attach earlier prices to the wrong security or erase the transition.

PYTHON / SYSTEM IMPLEMENTATION

Use surrogate instrument IDs and effective-dated mapping tables for vendor IDs, symbols, MICs, contracts, issuer, currency, and status. Enforce non-overlapping validity ranges.

CONFUSIONS TO AVOID

A permanent identifier is not always economic continuity: a futures contract expires, an ADR ratio changes, and a security can become non-tradable while its issuer survives.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The security master is a dependency of every normalized event; ingestion quarantines rows whose identity cannot be resolved at the event time.

RELATED CONCEPTS AND QUICK RECALL

Related: security master, symbology, effective dating, contract roll. Recall task: Describe a safe point-in-time join from vendor ticker to internal instrument ID.

Fundamentals, estimates, and data vintages

BEGINNER

ONE-SENTENCE MEANING

Fundamental and macro records have observation periods, publication times, revisions, and vendor arrival times that must be modeled separately.

MEMORY JOGGER

The number for a past quarter is not necessarily the number investors saw then. Research needs the vintage, not only the latest value.

WHY IT MATTERS IN QUANT RESEARCH

Value, quality, surprise, and macro strategies can leak revised statements or restated history unless features replay the original availability path.

FORMULA INTUITION

A point-in-time feature uses the latest record with `period_end <= cutoff` logic and `publish_time <= decision_time`, plus revision sequence when several vintages exist.

SIMPLE MARKET EXAMPLE

A company reports earnings, later restates revenue, and the vendor updates all history. A latest-snapshot database gives the corrected value to a backtest before the restatement existed.

PYTHON / SYSTEM IMPLEMENTATION

Store period, fiscal convention, announce time, effective time, vendor arrival, revision ID, preliminary/final status, units, currency, scaling, and source document.

CONFUSIONS TO AVOID

Quarter labels are not universal: fiscal calendars differ, trailing periods overlap, and vendor standardized fields can change definition. Filing date alone may not equal machine availability.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The point-in-time warehouse exposes as-of queries and feature snapshots; latest-value views are allowed for current reporting but barred from historical model training.

RELATED CONCEPTS AND QUICK RECALL

Related: revision, restatement, fiscal period, as-of query. Recall task: Which four dates would you preserve for an earnings fact, and which one controls historical feature eligibility?

Alternative and derived market data

BEGINNER

ONE-SENTENCE MEANING

Alternative data converts nontraditional observations into market-linked records, while derived data applies a model or transformation to primary inputs.

MEMORY JOGGER

A derived field inherits every weakness of its source plus the assumptions of its producer. Vendor scores are models, not raw facts.

WHY IT MATTERS IN QUANT RESEARCH

News, web traffic, satellite, card, sentiment, and flow estimates may add information, but coverage drift, entity mapping, revisions, and legal constraints can dominate apparent alpha.

FORMULA INTUITION

Coverage-adjusted rate may be $x_t = \text{sum observed activity} / \text{eligible exposure}$; denominator changes must be separated from true activity changes.

SIMPLE MARKET EXAMPLE

A sentiment vendor improves its classifier and reprocesses five years of articles. A latest historical dump shows a stable signal that never existed in real time.

PYTHON / SYSTEM IMPLEMENTATION

Capture raw payload references, collection window, arrival, entity link, coverage denominator, model version, reprocessing policy, license, and permitted use. Maintain vintage snapshots.

CONFUSIONS TO AVOID

High feature count and weak provenance invite selection bias. A vendor backtest can reflect future entity mappings, survivorship, or algorithm versions unavailable to subscribers then.

WHERE THIS FITS IN THE RESEARCH SYSTEM

An alternative-data quarantine layer separates legal review, identity resolution, raw archiving, feature derivation, and point-in-time publication before the research catalog exposes it.

RELATED CONCEPTS AND QUICK RECALL

Related: entity resolution, coverage bias, vendor model, data licensing. Recall task: Name three lineage fields needed before a vendor sentiment score can be tested causally.

Long, wide, and nested representations

BEGINNER

ONE-SENTENCE MEANING

Long tables store one observation or field per row, wide tables place many instruments or fields across columns, and nested records keep related structures together.

MEMORY JOGGER

Choose the shape for the operation: long for scalable filtering and joins, wide for matrix math, nested for event payloads that belong together.

WHY IT MATTERS IN QUANT RESEARCH

Research code often pivots long market events into wide matrices, but universes and missingness make wide schemas unstable. Storing only wide arrays can lose field-level lineage.

FORMULA INTUITION

A long key may be (time, instrument, field)->value; a wide matrix is $X[\text{time}, \text{instrument}]$. Pivot requires unique keys and an explicit aggregation rule.

SIMPLE MARKET EXAMPLE

Five assets over three days form fifteen long rows or a 3x5 return matrix. A duplicate asset-day makes the pivot ambiguous unless a correction or aggregation policy resolves it.

PYTHON / SYSTEM IMPLEMENTATION

Keep normalized long tables as durable interfaces; create wide arrays at bounded research steps with a stored column order. Use nested types for order-book levels or structured metadata.

CONFUSIONS TO AVOID

Wide is not inherently faster for every task, and long is not automatically tidy if rows mix incompatible grains. Nested convenience can make predicate filtering and schema evolution harder.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The dataset builder publishes a long canonical table plus materialized matrices whose universe order, missing policy, and source snapshot are recorded with the artifact.

RELATED CONCEPTS AND QUICK RECALL

Related: pivot, tidy data, sparse matrix, nested schema. Recall task: What must be true about keys before a long return table can safely pivot to a matrix?

Grain and primary-key design

BEGINNER

ONE-SENTENCE MEANING

Dataset grain states exactly what one row represents, and the primary key identifies that row uniquely within the contract.

MEMORY JOGGER

Say one row per what before naming columns. Mixed grain is the source of many duplicated joins and inflated statistics.

WHY IT MATTERS IN QUANT RESEARCH

A trade, quote update, minute bar, daily instrument fact, and portfolio snapshot require different keys. Joining two grains without aggregation can multiply records and bias volume or exposure.

FORMULA INTUITION

If table A has n keys and table B has m matches per key, an uncontrolled join can create $n*m$ rows; row-count and key-cardinality checks reveal the explosion.

SIMPLE MARKET EXAMPLE

Joining daily prices to every intraday quote repeats the daily close thousands of times. Summing after the join produces nonsense even though all individual values look valid.

PYTHON / SYSTEM IMPLEMENTATION

Declare key columns, uniqueness, parent grain, aggregation rules, and allowed relationship cardinality. Assert one-to-one, many-to-one, or bounded one-to-many in code.

CONFUSIONS TO AVOID

A timestamp alone is rarely a primary key; multiple venues and events share times. A surrogate row ID does not fix unclear economic grain.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The catalog exposes grain and key metadata, while dataset builders fail on unexpected duplicates, cardinality changes, or mixed observation types.

RELATED CONCEPTS AND QUICK RECALL

Related: primary key, cardinality, normalization, join explosion. Recall task: Define the grain and key for a consolidated one-minute instrument bar table.

Append-only event logs and state tables

BEGINNER

ONE-SENTENCE MEANING

An event log preserves changes in order, while a state table stores the latest or point-in-time result after applying those changes.

MEMORY JOGGER

Events explain how you arrived; state tells you where you are. Reliable systems often need both.

WHY IT MATTERS IN QUANT RESEARCH

Trades, quote messages, reference changes, and corrections are naturally event-based. Researchers often prefer state snapshots, but snapshots without lineage are hard to audit or repair.

FORMULA INTUITION

State S_t is a fold over ordered events: $S_t = \text{reduce}(\text{apply}, S_0, e_1 \dots e_t)$. Determinism requires a total order and idempotent event identity.

SIMPLE MARKET EXAMPLE

An order-book add followed by cancel produces no resting order. A current snapshot hides the lifecycle, while the log supports queue, cancellation, and reconstruction research.

PYTHON / SYSTEM IMPLEMENTATION

Store immutable raw events, normalized events, checkpoints, and derived state. Rebuild from the last verified checkpoint and compare hashes against live state.

CONFUSIONS TO AVOID

Append-only does not mean errors are ignored; corrections become new events. Mutating history destroys evidence and makes downstream snapshots impossible to reproduce.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The event store is the audit source, state services support low-latency consumers, and periodic reconciliation proves that both representations agree.

RELATED CONCEPTS AND QUICK RECALL

Related: event sourcing, checkpoint, materialized view, replay. Recall task: Why is a correction event safer than overwriting the original market record?

Trading calendars, sessions, and time zones

BEGINNER

ONE-SENTENCE MEANING

A market calendar defines valid sessions, opens, closes, auctions, holidays, early closes, and timezone transitions for a venue or instrument.

MEMORY JOGGER

Clock time without a calendar cannot tell you whether a bar should exist or whether two markets were simultaneously open.

WHY IT MATTERS IN QUANT RESEARCH

Returns, daily boundaries, overnight gaps, seasonality, and global joins depend on calendar logic. Daylight-saving changes can shift UTC overlap even when local opens remain fixed.

FORMULA INTUITION

A session key maps timestamp t to a trading label through venue-specific intervals rather than calendar date alone.

SIMPLE MARKET EXAMPLE

A U.S. equity early close has fewer minute bars; forcing a normal session inserts synthetic emptiness. A London-New York overlap changes in weeks when daylight transitions differ.

PYTHON / SYSTEM IMPLEMENTATION

Use a versioned exchange calendar service and store both UTC event time and local session labels. Test holidays, half-days, auctions, overnight futures, and daylight boundaries.

CONFUSIONS TO AVOID

Midnight UTC is not a universal daily cutoff. Futures sessions can cross calendar days, and corporate events may use issuer-local dates rather than trading sessions.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The calendar service is shared by ingestion, bar construction, feature windows, validation splits, and execution simulation so every layer agrees on elapsed market time.

RELATED CONCEPTS AND QUICK RECALL

Related: session, holiday calendar, timezone, trading-day label. Recall task: Give one example where calendar date and trading session label differ.

Resampling and interval boundaries

BEGINNER

ONE-SENTENCE MEANING

Resampling converts observations from one clock or frequency to another using declared bins, labels, closures, and aggregation functions.

MEMORY JOGGER

A five-minute return is not just every fifth row. It is a transformation of a precisely bounded event interval.

WHY IT MATTERS IN QUANT RESEARCH

Features and targets change with sampling frequency. Boundary mistakes create off-by-one leakage, duplicated events, or missing open and close auctions.

FORMULA INTUITION

For bins $[b_k, b_{(k+1)})$, each event belongs once when left-closed and right-open; label and close conventions must match joins and action times.

SIMPLE MARKET EXAMPLE

A trade exactly at 10:05 belongs either to the bar ending 10:05 or starting 10:05, not both. Different libraries have different defaults.

PYTHON / SYSTEM IMPLEMENTATION

Centralize resampling policy and unit-test boundary timestamps, empty intervals, late events, session transitions, and partial bins. Persist interval start and end, not only a label.

CONFUSIONS TO AVOID

Downsampling discards path information; upsampling creates a grid but no new information. Forward-filled values need an observation-age field.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature layer consumes canonical interval objects so price, volume, labels, and costs share identical boundary semantics.

RELATED CONCEPTS AND QUICK RECALL

Related: bin closure, interval label, downsampling, upsampling. Recall task: Where should an event exactly on a boundary go under a left-closed convention?

Irregular sampling and event-time clocks

BEGINNER

ONE-SENTENCE MEANING

Irregular data arrives when market activity occurs, so elapsed time between observations varies and silence can itself be informative.

MEMORY JOGGER

Row lag and clock lag are different. Ten trades can happen in a millisecond or over an hour.

WHY IT MATTERS IN QUANT RESEARCH

Tick models, volatility estimates, and decay calculations need actual durations. Regularizing too early can hide activity bursts, staleness, or no-trade periods.

FORMULA INTUITION

Duration $\Delta t_i = t_i - t_{(i-1)}$ should travel with the observation; intensity can be estimated as events per unit clock time over a declared window.

SIMPLE MARKET EXAMPLE

A liquid asset prints hundreds of times during a news burst while an illiquid name remains unchanged. Equal-row rolling windows represent radically different market exposure.

PYTHON / SYSTEM IMPLEMENTATION

Retain event time and duration features; choose calendar-time, tick-time, volume-time, or dollar-time sampling based on the decision. Model censoring when no event occurs.

CONFUSIONS TO AVOID

Treating unchanged last price as a fresh observation understates uncertainty. Event-time normalization also complicates cross-asset synchronization.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The sampling layer produces multiple governed clocks from the same immutable events, allowing later comparison without reinterpreting the raw feed.

RELATED CONCEPTS AND QUICK RECALL

Related: duration, intensity, business time, censoring. Recall task: Why can a 100-trade rolling window represent different economic horizons across assets?

Point-in-time and as-of joins

BEGINNER

ONE-SENTENCE MEANING

An as-of join matches each decision row to the latest eligible record known before its cutoff, subject to identity, direction, tolerance, and validity rules.

MEMORY JOGGER

Match backward in knowledge time, then verify the matched value was effective for the economic period you intend.

WHY IT MATTERS IN QUANT RESEARCH

Quotes, fundamentals, borrow, risk limits, and features update asynchronously. Ordinary equality joins drop useful observations, while unconstrained forward joins leak future values.

FORMULA INTUITION

For decision time t , choose record j maximizing available_time_j such that $\text{available_time_j} \leq t$ and $t - \text{available_time_j} \leq \text{tolerance}$.

SIMPLE MARKET EXAMPLE

A 10:00 signal uses the 09:59:58 quote if no newer quote arrived and the tolerance allows two seconds. A 10:00:01 quote cannot be matched backward to the decision.

PYTHON / SYSTEM IMPLEMENTATION

Sort by entity and time, specify direction and tolerance, preserve matched timestamps and age, and test future perturbation. For effective-dated facts, require both availability and validity.

CONFUSIONS TO AVOID

Latest-before is not always valid forever. Stale values, revoked estimates, session boundaries, and entity changes require explicit expiry.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The point-in-time join library emits match age, source ID, cutoff, and rejection reason so feature lineage and missingness remain auditable.

RELATED CONCEPTS AND QUICK RECALL

Related: as-of join, validity interval, staleness, leakage. Recall task: Write the eligibility predicate for joining a quote to a historical decision row.

Units, currencies, and price conventions

BEGINNER

ONE-SENTENCE MEANING

Every numeric field needs units and conventions that define scale, currency, multiplier, sign, and whether values are raw, adjusted, quoted, or normalized.

MEMORY JOGGER

Numbers with different units can be joined and summed by software while remaining economically meaningless.

WHY IT MATTERS IN QUANT RESEARCH

Futures multipliers, bond price points, FX quote direction, crypto base/quote pairs, and share-lot conventions affect returns, notional, P&L, and risk.

FORMULA INTUITION

Notional may be $\text{price} \times \text{quantity} \times \text{contract_multiplier}$; FX conversion requires a rate with declared base, quote, timestamp, and conversion direction.

SIMPLE MARKET EXAMPLE

A futures price move of one index point is not one dollar when the contract multiplier is fifty. Reversing EUR/USD to USD/EUR changes nonlinear return arithmetic.

PYTHON / SYSTEM IMPLEMENTATION

Attach unit metadata to schemas and normalize only through versioned transformations. Assert compatible dimensions before aggregation and store source and converted values together.

CONFUSIONS TO AVOID

Standardizing to z-scores removes units for a model but not for target interpretation or execution. Percent, decimal return, basis points, and price points are frequently confused.

WHERE THIS FITS IN THE RESEARCH SYSTEM

A unit service supplies contract terms and FX conversions to bars, positions, costs, and risk; data tests reconcile normalized notional to source statements.

RELATED CONCEPTS AND QUICK RECALL

Related: contract multiplier, FX conversion, basis point, quote convention. Recall task: Compute the dollar effect of a one-point move only after naming which metadata fields?

Row-oriented and column-oriented storage

BEGINNER

ONE-SENTENCE MEANING

Row-oriented systems keep fields of one record together, while column-oriented systems group values of the same field for efficient scans and compression.

MEMORY JOGGER

Rows favor point updates and complete-record lookup; columns favor analytical scans over a subset of fields.

WHY IT MATTERS IN QUANT RESEARCH

Research frequently reads a few columns across millions of bars, making columnar storage effective. Order and state services may need low-latency keyed row access.

FORMULA INTUITION

I/O cost depends on rows touched, columns projected, encoding, and selectivity; no orientation is universally fastest.

SIMPLE MARKET EXAMPLE

A factor study reading time, instrument, close, and volume from a hundred-field dataset avoids most bytes in a columnar file. Fetching one full instrument snapshot may favor a row store.

PYTHON / SYSTEM IMPLEMENTATION

Benchmark representative queries, not synthetic slogans. Keep transaction workloads and analytical workloads separable, with governed replication between them.

CONFUSIONS TO AVOID

Columnar files do not provide database transactions by themselves, and a row database can still use column indexes or analytical replicas. Layout is one layer of the system.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Operational stores serve current state; immutable columnar partitions serve historical research; reconciliation confirms both encode the same normalized events.

RELATED CONCEPTS AND QUICK RECALL

Related: OLTP, OLAP, predicate pushdown, projection. Recall task: Choose an orientation for historical factor scans and justify the opposite choice for a live security-master lookup.

CSV, Parquet, and Arrow

BEGINNER

ONE-SENTENCE MEANING

CSV is a portable text interchange format, Parquet is a typed compressed columnar file, and Arrow is an in-memory columnar representation for efficient interchange.

MEMORY JOGGER

File extension is not a data contract. Type fidelity, nulls, metadata, partitioning, and reader behavior determine reliability.

WHY IT MATTERS IN QUANT RESEARCH

CSV is inspectable but loses types and scans slowly; Parquet supports projection and predicate pushdown; Arrow reduces serialization between compatible processes.

FORMULA INTUITION

Storage size and read time roughly depend on encoding plus entropy; repeated typed values compress far better than textual duplication.

SIMPLE MARKET EXAMPLE

A timestamp column in CSV may be parsed differently by locale or library. Parquet preserves an explicit timestamp type, but timezone and unit still require contract metadata.

PYTHON / SYSTEM IMPLEMENTATION

Use Parquet for durable analytical datasets, Arrow for in-memory boundaries, and CSV only for bounded exchange or audit views. Validate schema fingerprints after every read.

CONFUSIONS TO AVOID

Parquet files can still disagree on field types across partitions. Arrow zero-copy claims do not remove ownership, lifetime, or compatibility concerns.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The dataset writer publishes schema, row count, statistics, checksums, partitions, and source snapshot alongside files; readers verify them before materialization.

RELATED CONCEPTS AND QUICK RECALL

Related: schema-on-read, columnar file, serialization, predicate pushdown. Recall task: Why can a directory of individually valid Parquet files still fail as one dataset?

Partitioning, clustering, and small files

BEGINNER

ONE-SENTENCE MEANING

Partitioning places data into directory or table segments using common filter keys, while clustering orders or groups values within those segments.

MEMORY JOGGER

Partition for coarse pruning, cluster for finer locality, and avoid creating more physical objects than the platform can manage efficiently.

WHY IT MATTERS IN QUANT RESEARCH

Date and venue partitions can reduce scans, but instrument-by-minute partitioning creates millions of tiny files and expensive metadata operations.

FORMULA INTUITION

Useful partition cardinality balances bytes per partition, write concurrency, retention, and common predicates; the optimum is workload-specific.

SIMPLE MARKET EXAMPLE

Daily Parquet partitions containing all instruments let a one-year market-wide scan prune by date. One file per symbol per minute overwhelms listing and scheduling overhead.

PYTHON / SYSTEM IMPLEMENTATION

Measure file-size distribution, partition count, pruning ratio, and query patterns. Compact small files deterministically while preserving snapshot lineage and atomic publication.

CONFUSIONS TO AVOID

A partition key is duplicated in paths and may be absent from file columns depending on the engine. Over-partitioning also makes late corrections and schema evolution harder.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The lake writer stages data, validates complete partitions, publishes a manifest atomically, and runs compaction without changing logical dataset identity.

RELATED CONCEPTS AND QUICK RECALL

Related: partition pruning, clustering, compaction, manifest. Recall task: What symptoms show that a market-data lake is over-partitioned?

Database choices for market data

BEGINNER

ONE-SENTENCE MEANING

Market-data platforms combine relational, time-series, object, key-value, and streaming systems because no single engine optimizes every access pattern.

MEMORY JOGGER

Choose from the query and correctness contract: point lookup, range scan, join, replay, current state, or low-latency fan-out.

WHY IT MATTERS IN QUANT RESEARCH

Relational stores provide constraints and joins; time-series engines optimize time filters; object stores scale immutable history; key-value stores serve current snapshots; logs distribute ordered events.

FORMULA INTUITION

Evaluate p95 latency, throughput, recovery point, consistency, retention, scan cost, concurrency, and operational burden under expected data volume.

SIMPLE MARKET EXAMPLE

A research lake can hold years of trades cheaply while a key-value cache serves the latest best quote. A relational catalog maintains instruments and dataset metadata.

PYTHON / SYSTEM IMPLEMENTATION

Document workload, failure model, ownership, backup, restore, and export path before selecting technology. Prototype with representative data and skew.

CONFUSIONS TO AVOID

Time-series branding does not guarantee point-in-time correctness or cheap cross-sectional joins. Moving systems can break event order and identifier mapping.

WHERE THIS FITS IN THE RESEARCH SYSTEM

A layered architecture assigns each engine a bounded role and reconciles data across boundaries using immutable event IDs and snapshot manifests.

RELATED CONCEPTS AND QUICK RECALL

Related: relational database, object store, stream log, cache. Recall task: Map four market-data workloads to appropriate storage roles and state the reconciliation key.

Compression, encoding, and precision

BEGINNER

ONE-SENTENCE MEANING

Compression reduces bytes by exploiting repeated structure, while encoding determines how values such as timestamps, decimals, enums, and nulls are physically represented.

MEMORY JOGGER

Efficient storage begins with correct types. Textual numbers and uncontrolled floating point waste space and can lose meaning.

WHY IT MATTERS IN QUANT RESEARCH

Delta-encoded timestamps, dictionary-coded symbols, and compressed numeric columns reduce I/O. Precision choices affect exact price reconstruction and key comparisons.

FORMULA INTUITION

Integer ticks can encode price as $\text{round}(\text{price}/\text{tick_size})$, while scaled integers represent fixed decimals without binary floating ambiguity.

SIMPLE MARKET EXAMPLE

Price 0.1 cannot be represented exactly in binary floating point. Repeated conversion can break equality checks at a tick boundary, while integer ticks remain exact.

PYTHON / SYSTEM IMPLEMENTATION

Use fixed-point or integer ticks where exactness matters, record scale and tick regime, and choose codecs from measured CPU/I/O tradeoffs. Verify round-trip values and null semantics.

CONFUSIONS TO AVOID

More compression is not always faster when CPU dominates, and lossy numeric compression is unsafe for audit or execution data unless explicitly bounded and separated.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The storage writer applies schema-driven encodings; dataset validation performs min/max, precision, checksum, and lossless round-trip tests before publication.

RELATED CONCEPTS AND QUICK RECALL

Related: fixed point, dictionary encoding, delta encoding, floating point. Recall task: Why is integer-tick storage useful for prices, and what metadata is required to decode it?

Schema evolution and compatibility

BEGINNER

ONE-SENTENCE MEANING

Schema evolution changes fields or meanings over time while preserving a governed path for old and new producers and consumers.

MEMORY JOGGER

Adding a nullable field is usually easier than changing the meaning of an existing one. Semantic changes deserve new versions even when types stay identical.

WHY IT MATTERS IN QUANT RESEARCH

Feed upgrades, vendor fields, precision, and identifiers evolve. Uncontrolled drift can silently coerce types, drop values, or make historical partitions unreadable together.

FORMULA INTUITION

Compatibility can be backward, forward, or full: new readers handling old data, old readers handling new data, or both.

SIMPLE MARKET EXAMPLE

A size field changes from shares to lots without a name change. The integer type remains valid, so type-only tests pass while liquidity estimates shift by a factor of one hundred.

PYTHON / SYSTEM IMPLEMENTATION

Version contracts, maintain transformation adapters, test representative old/new payloads, and publish deprecation windows. Compare both structural fingerprints and semantic metadata.

CONFUSIONS TO AVOID

Renaming a field is often drop-plus-add to machines. Default values can mask missing producer support, and unioning partitions can widen types unexpectedly.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The registry blocks incompatible publication, while consumers declare accepted contract ranges and observability reports which versions remain active.

RELATED CONCEPTS AND QUICK RECALL

Related: backward compatibility, migration, semantic versioning, adapter. Recall task: Give an example of a breaking semantic change that a structural schema checker would miss.

Snapshots, versions, and reproducible datasets

BEGINNER

ONE-SENTENCE MEANING

A dataset snapshot is an immutable, addressable view of exact files, schema, partitions, and source lineage used at a point in research or production.

MEMORY JOGGER

A query against a mutable table is not a reproducible input. Record the data state, not only the SQL text.

WHY IT MATTERS IN QUANT RESEARCH

Vendor corrections, late events, corporate actions, and compaction change physical data. Snapshots let experiments, audits, and incident replay return to the same logical rows.

FORMULA INTUITION

A snapshot manifest can hash ordered file paths plus file checksums and schema ID, producing a stable content address.

SIMPLE MARKET EXAMPLE

A strategy trained on Monday cannot be reproduced Friday after backfilled trades unless the Monday snapshot or equivalent change log remains accessible.

PYTHON / SYSTEM IMPLEMENTATION

Publish immutable manifests atomically; include source cutoffs, row counts, partition stats, schema, transforms, and retention. Bind experiment IDs to snapshot IDs.

CONFUSIONS TO AVOID

Copying a directory name is not versioning if files can change in place. Physical compaction may preserve logical content but requires a new manifest with equivalence proof.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The experiment registry, feature store, and model registry share snapshot identifiers so every result can be replayed from raw events through decisions.

RELATED CONCEPTS AND QUICK RECALL

Related: time travel, manifest, content hash, reproducibility. Recall task: What must a snapshot manifest include beyond a timestamp?

Metadata catalogs and lineage graphs

BEGINNER

ONE-SENTENCE MEANING

A data catalog describes datasets, contracts, owners, quality, access, and dependencies; lineage records how outputs were derived from exact inputs and code.

MEMORY JOGGER

Metadata answers what this is, where it came from, who owns it, and what breaks if it changes.

WHY IT MATTERS IN QUANT RESEARCH

Quant systems contain many near-identical prices, returns, universes, and features. Searchable lineage prevents accidental mixing and speeds incident impact analysis.

FORMULA INTUITION

A lineage edge can be modeled `output_artifact <- transformation(code_hash, parameters) <- input_artifact`, forming a directed acyclic graph for a completed run.

SIMPLE MARKET EXAMPLE

A return anomaly is traced from a model feature to adjusted bars, a corporate-action factor, normalized trades, and one vendor correction through stored artifact IDs.

PYTHON / SYSTEM IMPLEMENTATION

Capture dataset purpose, grain, schema, owner, SLA, sensitivity, freshness, tests, source, code hash, parameters, and downstream consumers automatically at publication.

CONFUSIONS TO AVOID

A catalog that depends on manual descriptions quickly becomes stale. Column-level lineage is useful but cannot replace semantic definitions or ownership.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The control plane uses lineage to block expired inputs, route alerts to owners, estimate blast radius, and reproduce the exact path behind a capital decision.

RELATED CONCEPTS AND QUICK RECALL

Related: data catalog, provenance, DAG, blast radius. Recall task: Trace the minimum lineage chain from a model score back to raw trade events.

Missingness, staleness, and halted markets

BEGINNER

ONE-SENTENCE MEANING

Missing data means an expected observation is absent; stale data means an old value is carried forward; a halt is a market state that may legitimately suppress updates.

MEMORY JOGGER

No new value can mean feed failure, illiquidity, closure, or halt. The reason matters more than the null itself.

WHY IT MATTERS IN QUANT RESEARCH

Features, marks, and risk estimates react differently to missing and stale inputs. Treating all gaps as zero or forward-filling indefinitely understates uncertainty and can trigger false trades.

FORMULA INTUITION

Age at decision is $t_{\text{decision}} - t_{\text{last_observed}}$; health thresholds should depend on instrument, session state, and data type.

SIMPLE MARKET EXAMPLE

A quote remains unchanged for twenty minutes. In a liquid open market it signals a broken feed; during a halt it is expected but not executable.

PYTHON / SYSTEM IMPLEMENTATION

Emit `observed_time`, `age`, `expected-update class`, `session status`, `halt flag`, `imputation method`, and `health tier`. Keep imputed values distinguishable from observations.

CONFUSIONS TO AVOID

A regular time grid makes absence look like data. Forward-filled prices should not create zero-return evidence, fresh liquidity, or additional sample size.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Health services combine feed, calendar, halt, and instrument metadata; downstream policies abstain, widen uncertainty, or switch sources based on reason-coded status.

RELATED CONCEPTS AND QUICK RECALL

Related: missingness, freshness, halt, imputation. Recall task: How would you distinguish an illiquid instrument from a dead quote feed?

Duplicates, corrections, and out-of-order events

BEGINNER

ONE-SENTENCE MEANING

Duplicate and reordered events arise from retries, multiple routes, clock differences, and late corrections, so ingestion needs explicit identity and ordering rules.

MEMORY JOGGER

Delivery can be repeated and delayed. Economic events should be applied exactly according to their identity and sequence, not packet arrival alone.

WHY IT MATTERS IN QUANT RESEARCH

Double-counted trades inflate volume; reordered book messages corrupt depth; discarded corrections preserve known errors. These faults can survive into polished bars.

FORMULA INTUITION

Deduplication uses a stable event key or source sequence; ordering may combine session, channel, sequence, and event time with a bounded lateness policy.

SIMPLE MARKET EXAMPLE

A consumer restarts and receives the last thousand messages again. Without idempotent event IDs, every trade and cancel is applied twice.

PYTHON / SYSTEM IMPLEMENTATION

Maintain raw delivery IDs and normalized economic IDs, detect gaps, buffer bounded disorder, and process corrections as linked events. Reconcile state after recovery.

CONFUSIONS TO AVOID

Equal payloads are not necessarily duplicates, and different payloads can correct the same event. Sorting all history later cannot repair live decisions made from corrupt state.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The ingestion layer publishes quality counters and sequence health; stateful consumers checkpoint offsets and can rebuild from raw events when reconciliation fails.

RELATED CONCEPTS AND QUICK RECALL

Related: idempotence, sequence gap, correction, late data. Recall task: Why do you need both delivery identity and economic event identity?

Bad ticks, outliers, and price validation

BEGINNER

ONE-SENTENCE MEANING

A bad tick is an erroneous market record, while an outlier is merely unusual; validation should classify evidence before altering values.

MEMORY JOGGER

Extreme does not mean wrong. Compare the event with quotes, venues, conditions, neighboring records, and corporate actions.

WHY IT MATTERS IN QUANT RESEARCH

A single impossible price can dominate returns, volatility, stops, and model fitting. Removing genuine crisis moves, however, erases the risk the system must learn.

FORMULA INTUITION

Diagnostics include relative move, distance from contemporaneous midpoint, cross-venue agreement, tick-size validity, and reversal speed under state-aware thresholds.

SIMPLE MARKET EXAMPLE

A trade at one-tenth the prevailing price with an immediate correction is likely erroneous; a similar move across venues after news may be genuine.

PYTHON / SYSTEM IMPLEMENTATION

Preserve raw records, attach reason-coded flags and confidence, and create policy-specific eligible views. Validate against reference terms, quote bands, venue conditions, and correction links.

CONFUSIONS TO AVOID

Winsorization is a modeling choice, not data repair. Static percentage thresholds fail across price levels, volatility regimes, splits, and illiquid instruments.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Quality flags flow into bars, features, model weights, and incident dashboards so consumers can exclude, downweight, or stress suspicious observations transparently.

RELATED CONCEPTS AND QUICK RECALL

Related: outlier, correction, robust statistics, quality flag. Recall task: What evidence would convince you an extreme trade is valid rather than erroneous?

Crossed, locked, and inconsistent books

BEGINNER

ONE-SENTENCE MEANING

A locked market has equal best bid and ask, while a crossed market has bid above ask; either may be real, transient, or a reconstruction symptom.

MEMORY JOGGER

Book geometry is a health signal. Do not automatically coerce impossible-looking states before checking clocks, venue scope, and eligibility.

WHY IT MATTERS IN QUANT RESEARCH

Spread features, midpoint marks, and execution models fail when book state is inconsistent. Crosses can expose asynchronous venue feeds, sequence gaps, or different condition filters.

FORMULA INTUITION

Spread $s=a-b$; $s=0$ is locked and $s<0$ is crossed. Duration and source decomposition help distinguish fleeting consolidation effects from persistent corruption.

SIMPLE MARKET EXAMPLE

Two venue feeds arrive with unequal delay, so an old ask from one venue overlaps a new bid from another. The consolidated book crosses although neither venue book is crossed at its source time.

PYTHON / SYSTEM IMPLEMENTATION

Track crossed duration, venues, sequence health, timestamp deltas, and eligibility. Preserve venue-native state and derive consolidated state through a versioned rule.

CONFUSIONS TO AVOID

Dropping every crossed sample biases stress and open/close periods. Using a negative spread in cost models is equally wrong without a reachable arbitrage path.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The market-state service marks trustworthy, degraded, and invalid books; routing and research choose separate policies while quality monitoring retains the original evidence.

RELATED CONCEPTS AND QUICK RECALL

Related: locked market, crossed market, consolidation latency, book health. Recall task: Name two causes of a consolidated cross that do not imply a risk-free executable arbitrage.

Universe membership and survivorship history

BEGINNER

ONE-SENTENCE MEANING

A point-in-time universe records which instruments were eligible at each decision date, including entrants, exits, delistings, and data availability.

MEMORY JOGGER

Today's survivors are not yesterday's opportunity set. Dead, acquired, and untradeable names still belong in historical evidence.

WHY IT MATTERS IN QUANT RESEARCH

Cross-sectional returns, factors, capacity, and failure rates are overstated when research uses a current constituent list or drops delisted securities.

FORMULA INTUITION

Eligibility U_t is a function of information available before t : listing status, liquidity history, price, borrow, and policy, each with explicit lags.

SIMPLE MARKET EXAMPLE

Backtesting an index strategy on current members excludes firms removed after poor performance. The resulting historical portfolio benefits from future membership knowledge.

PYTHON / SYSTEM IMPLEMENTATION

Store effective-dated membership with add/remove reason, announcement, effective time, source vintage, and eligibility inputs. Recompute dynamic filters inside each historical cutoff.

CONFUSIONS TO AVOID

Point-in-time membership does not solve every bias: vendor coverage can begin late, delisting returns can be missing, and liquidity filters can use future volume if not lagged.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The universe service publishes immutable daily membership snapshots consumed by prices, features, labels, validation, and portfolio construction.

RELATED CONCEPTS AND QUICK RECALL

Related: survivorship bias, delisting return, constituent history, eligibility. Recall task: What future information enters when you backtest with today's index constituents?

Data contracts and quality tests

BEGINNER

ONE-SENTENCE MEANING

Data quality tests turn a dataset contract into executable checks for schema, keys, ranges, timing, completeness, consistency, and reconciliation.

MEMORY JOGGER

Test invariants close to the producer and decision-sensitive expectations close to the consumer.

WHY IT MATTERS IN QUANT RESEARCH

A dataset can be syntactically valid yet unusable for a strategy because of stale quotes, missing sessions, changed units, or late revisions. Layered tests catch both structural and economic faults.

FORMULA INTUITION

Quality dimensions include completeness, uniqueness, validity, timeliness, consistency, and accuracy against an independent reference where available.

SIMPLE MARKET EXAMPLE

A feed remains 100% non-null after size changes from shares to lots. A unit-reconciliation test detects the discontinuity that a null check misses.

PYTHON / SYSTEM IMPLEMENTATION

Version test suites with the contract; use hard gates for identity and impossible states, warning bands for distribution drift, and sampled reconciliations for external truth.

CONFUSIONS TO AVOID

Thresholds tuned after incidents merely describe the past. Excessive brittle tests create alert fatigue, while broad averages hide instrument and venue cohorts.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Publication requires a signed test report; the catalog exposes health, exceptions, owner, and waiver expiry to every downstream consumer.

RELATED CONCEPTS AND QUICK RECALL

Related: invariant, reconciliation, SLA, quality dimension. Recall task: Design one hard gate, one drift warning, and one reconciliation check for a quote dataset.

Batch and streaming pipelines

BEGINNER

ONE-SENTENCE MEANING

Batch pipelines process bounded datasets on a schedule, while streaming pipelines continuously transform unbounded events with state and lateness rules.

MEMORY JOGGER

Batch offers a complete window; streaming offers low delay. Both need the same semantic contract if their outputs are compared.

WHY IT MATTERS IN QUANT RESEARCH

Research often relies on batch history while live systems use streams. Divergent code paths create training-serving skew in bars, features, and quality decisions.

FORMULA INTUITION

Streaming windows require event-time boundaries, watermark or lateness policy, state retention, and update semantics; batch windows still need identical inclusion rules.

SIMPLE MARKET EXAMPLE

A live one-minute bar publishes at the boundary, then receives a late trade. The system must update, retract, or ignore according to policy; overnight batch should reproduce that chosen final state.

PYTHON / SYSTEM IMPLEMENTATION

Share transformation libraries and golden event fixtures. Compare stream snapshots with independent batch recomputation and publish correction events when allowed.

CONFUSIONS TO AVOID

Streaming is not automatically real time if sources are delayed, and batch is not inherently correct if it uses latest revised data. Operational complexity can exceed the latency benefit.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The platform separates immutable event ingestion, stateful live views, finalized analytical partitions, and reconciliation so each latency tier has explicit guarantees.

RELATED CONCEPTS AND QUICK RECALL

Related: watermark, window, training-serving skew, reconciliation. Recall task: What should happen when a late trade arrives after a live bar was published?

Idempotence and delivery guarantees

BEGINNER

ONE-SENTENCE MEANING

An idempotent operation can be repeated without changing the final logical result, making retries safe under failures and uncertain acknowledgments.

MEMORY JOGGER

Exactly-once business outcomes usually come from at-least-once delivery plus stable identities, transactions, and deterministic state updates.

WHY IT MATTERS IN QUANT RESEARCH

Market feeds and distributed pipelines retry, restart, and replay. Without idempotence, duplicated bars, factors, orders, or positions can appear during recovery.

FORMULA INTUITION

For event e and state transition f , idempotence requires $f(f(S,e),e)=f(S,e)$ for the same event identity.

SIMPLE MARKET EXAMPLE

A bar partition upload succeeds but the acknowledgment is lost. A retry should verify content and publish the same manifest, not append a second logical partition.

PYTHON / SYSTEM IMPLEMENTATION

Use deterministic keys, deduplication state, transactional output or compare-and-swap manifests, checkpointed offsets, and replay tests across crash points.

CONFUSIONS TO AVOID

Transport exactly-once labels do not guarantee end-to-end semantics when external databases, files, or side effects lack the same transaction boundary.

WHERE THIS FITS IN THE RESEARCH SYSTEM

Every pipeline stage exposes replay-safe contracts; incident recovery starts from verified checkpoints and reconciles content hashes before resuming publication.

RELATED CONCEPTS AND QUICK RECALL

Related: at-least-once, deduplication, transaction, checkpoint. Recall task: Explain how a stable event ID converts repeated delivery into one logical update.

Feature-ready interfaces and leakage barriers

BEGINNER

ONE-SENTENCE MEANING

A feature-ready interface publishes normalized, point-in-time market data with stable identity, timing, units, quality, and snapshot lineage for research and serving.

MEMORY JOGGER

The feature layer should not rediscover what a price or timestamp means. Upstream contracts remove ambiguity; downstream transforms still obey causal cutoffs.

WHY IT MATTERS IN QUANT RESEARCH

Shared interfaces reduce duplicated cleaning and training-serving skew, while leakage barriers prevent latest-adjusted or revised values from entering historical examples.

FORMULA INTUITION

A feature row should bind entity, event/effective time, available time, value, quality, transform version, and source snapshot to the model observation.

SIMPLE MARKET EXAMPLE

A rolling return uses adjusted bars and a point-in-time universe. If live serving uses raw prices or different session boundaries, the same feature name produces different economics.

PYTHON / SYSTEM IMPLEMENTATION

Provide as-of APIs, batch snapshots, freshness and quality columns, unit metadata, and deterministic transforms. Test future mutation and online/offline parity.

CONFUSIONS TO AVOID

Centralization can spread one bad definition widely, so ownership, versioning, canaries, and rollback matter. Feature stores do not automatically prevent label or split leakage.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The feature platform consumes governed market datasets and publishes immutable training sets plus low-latency values under one contract and parity test suite.

RELATED CONCEPTS AND QUICK RECALL

Related: feature store, offline-online parity, cutoff, training set. Recall task: Which fields must accompany a feature value to prove it was legal at a decision time?

Observability, reconciliation, and fallback

BEGINNER

ONE-SENTENCE MEANING

Market-data observability measures whether feeds and derived datasets are complete, timely, consistent, and fit for specific decisions, with tested degraded modes.

MEMORY JOGGER

Monitor both the pipes and the economics: bytes can flow while prices, units, coverage, or venue mix are wrong.

WHY IT MATTERS IN QUANT RESEARCH

Fast detection limits corrupted research and live exposure. Reconciliation against alternate sources or independent recomputation distinguishes source failure from transformation failure.

FORMULA INTUITION

Health can combine freshness, sequence gaps, coverage, cross-source deviation, correction rate, test failures, and consumer impact into tiered policy rather than one opaque score.

SIMPLE MARKET EXAMPLE

A primary quote feed remains timely but loses one venue. Aggregate spread and depth shift subtly; venue-coverage monitoring catches the defect before signal drift alone would.

PYTHON / SYSTEM IMPLEMENTATION

Build dashboards by source, venue, instrument cohort, and stage; retain raw samples; define warning, degraded, stop, fallback, and recovery criteria; rehearse source switches.

CONFUSIONS TO AVOID

A fallback with different timestamps, fields, or adjustment policy can be worse than stopping. Automatic recovery without backfill and state reconciliation preserves hidden gaps.

WHERE THIS FITS IN THE RESEARCH SYSTEM

The control plane attaches health to every dataset and blocks or reroutes consumers according to declared tolerances, while incident replay verifies the recovered history.

RELATED CONCEPTS AND QUICK RECALL

Related: observability, reconciliation, degraded mode, rollback. Recall task: Draft a three-tier response for quote freshness and venue-coverage deterioration.

Layered market-data architecture

BEGINNER

DESIGN PRINCIPLE

Preserve immutable evidence, derive governed state, and publish decision-specific interfaces. Each arrow carries contract, timing, quality, and lineage metadata.

RAW FEEDS

Trades | quotes | books | actions | reference | fundamentals

NORMALIZED EVENT LOG

Stable identity | source clocks | typed units | corrections

STATE AND DERIVATION

Books | bars | adjustments | universes | as-of facts

IMMUTABLE SNAPSHOTS

Parquet partitions | manifests | schemas | quality reports

RESEARCH AND SERVING

Matrices | features | backtests | APIs | live views

CONTROL PLANE

Catalog | lineage | health | fallback | rollback | ownership

FAILURE CONTAINMENT

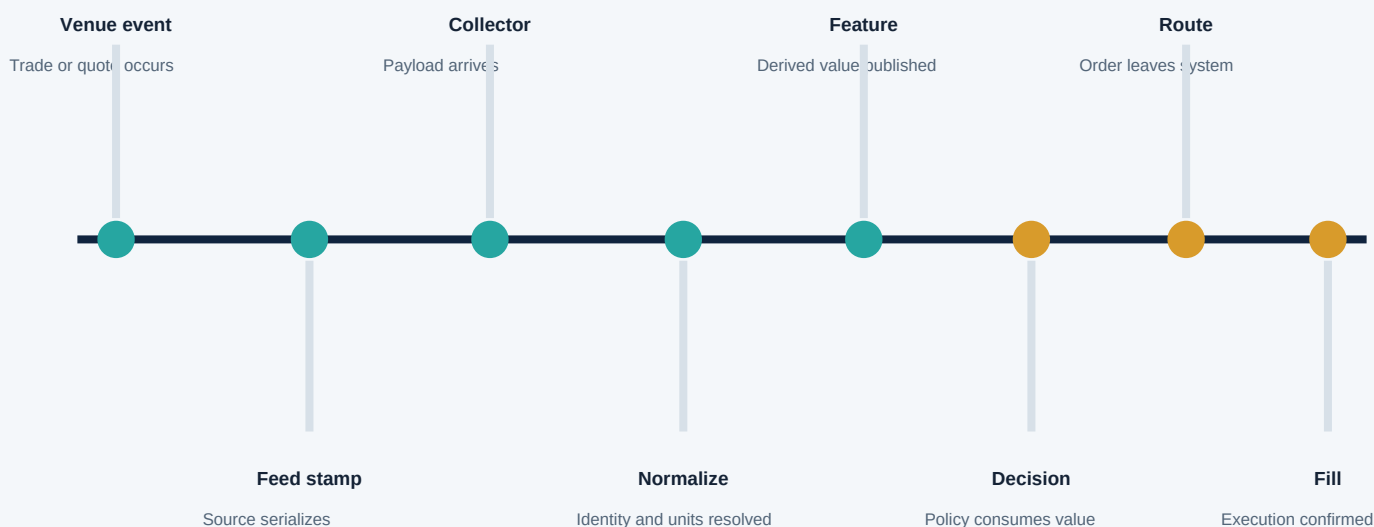
Consumers can stop or fall back without mutating raw evidence; repaired derivatives are published as new snapshots.

The full market-data knowledge clock

BEGINNER

READING RULE

The market can move before your system knows it. Historical replay must respect the first timestamp at which the exact normalized value was usable by the consuming decision.



CAUSAL ELIGIBILITY

A feature at the decision can use only records whose normalized publication precedes the decision cutoff. Event time reconstructs market order; arrival and publish time reconstruct system knowledge.

LATENCY ATTRIBUTION

Separating clocks lets operations locate network, normalization, feature, policy, routing, and venue delay instead of blaming one aggregate latency number.

AUDIT PAYLOAD

Persist all clocks, clock source, synchronization status, sequence, dataset snapshot, feature version, decision ID, order ID, and fill ID.

Order-book depth and executable cost

BEGINNER

SCENARIO

A buy order must consume asks from best to worse prices. The displayed top level is a starting point, not the cost for an order larger than that queue.

BIDS

SPREAD 0.01

ASKS

150 @ 100.00

120 @ 100.01

220 @ 99.99

180 @ 100.02

310 @ 99.98

260 @ 100.03

460 @ 99.96

500 @ 100.05

WORKED SWEEP

A 250-share market buy takes 120 at 100.01 and 130 at 100.02. Average execution is $(120 \times 100.01 + 130 \times 100.02) / 250 = 100.0152$, before fees and hidden liquidity.

STATE CAVEAT

Displayed size can cancel, replenish, or lose priority before arrival. A replay needs message order, latency, queue model, hidden-liquidity assumptions, and venue routing.

RECALL

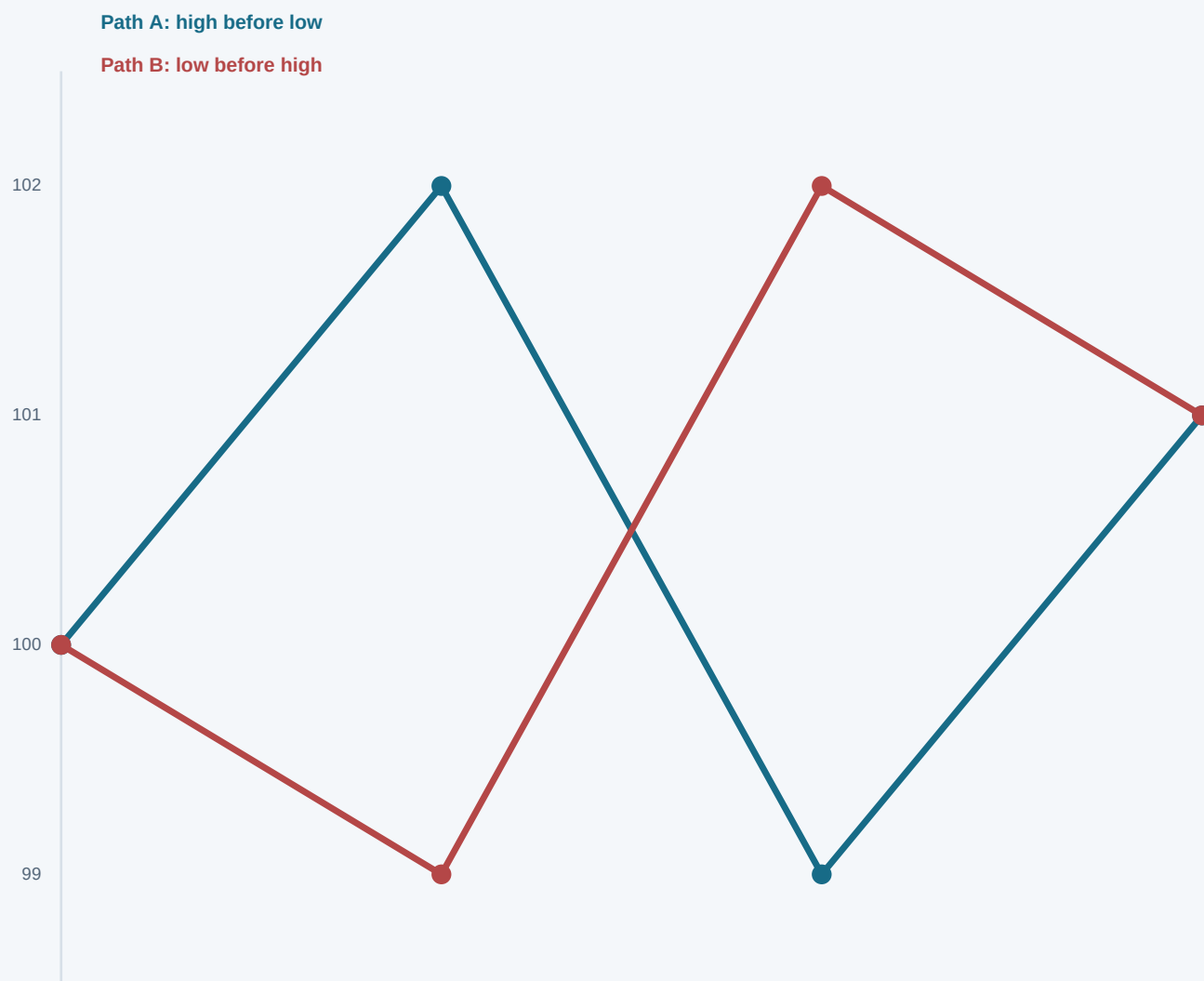
Explain why L1 spread, L2 sweep cost, and L3 queue simulation answer progressively different execution questions.

One bar, many possible event paths

BEGINNER

CORE LESSON

OHLCV preserves boundary and extremes but loses the order of events. Stops, limits, spreads, and market impact cannot be reconstructed uniquely from one bar.



IDENTICAL SUMMARY

Both paths have $O=100$, $H=102$, $L=99$, $C=101$. A stop at 99.5 and a target at 101.5 produce different trade outcomes because event order differs.

RESEARCH RULE

Use event data or conservative intrabar assumptions when path order affects labels, fills, stops, or drawdowns. Document the ambiguity rather than selecting the favorable sequence.

Structure selection by workload

BEGINNER

DATA OBJECT	DEFAULT STRUCTURE	PRIMARY STRENGTH	MAIN RISK
Raw feed	Append-only log	Replay and audit	Needs ordering and retention
Current quote	Key-value state	Low-latency lookup	Loses path unless logged
Historical bars	Columnar partitions	Range scans and compression	Late correction workflow
Security master	Relational temporal table	Constraints and joins	Effective-date complexity
Return matrix	Wide array	Linear algebra	Unstable universe order
Feature history	Long point-in-time table	As-of joins and lineage	Pivot cost
Order book	Event log plus checkpoints	Queue replay and restart	Stateful reconstruction
Experiment input	Immutable snapshot	Exact reproducibility	Retention and manifests

SELECTION DISCIPLINE

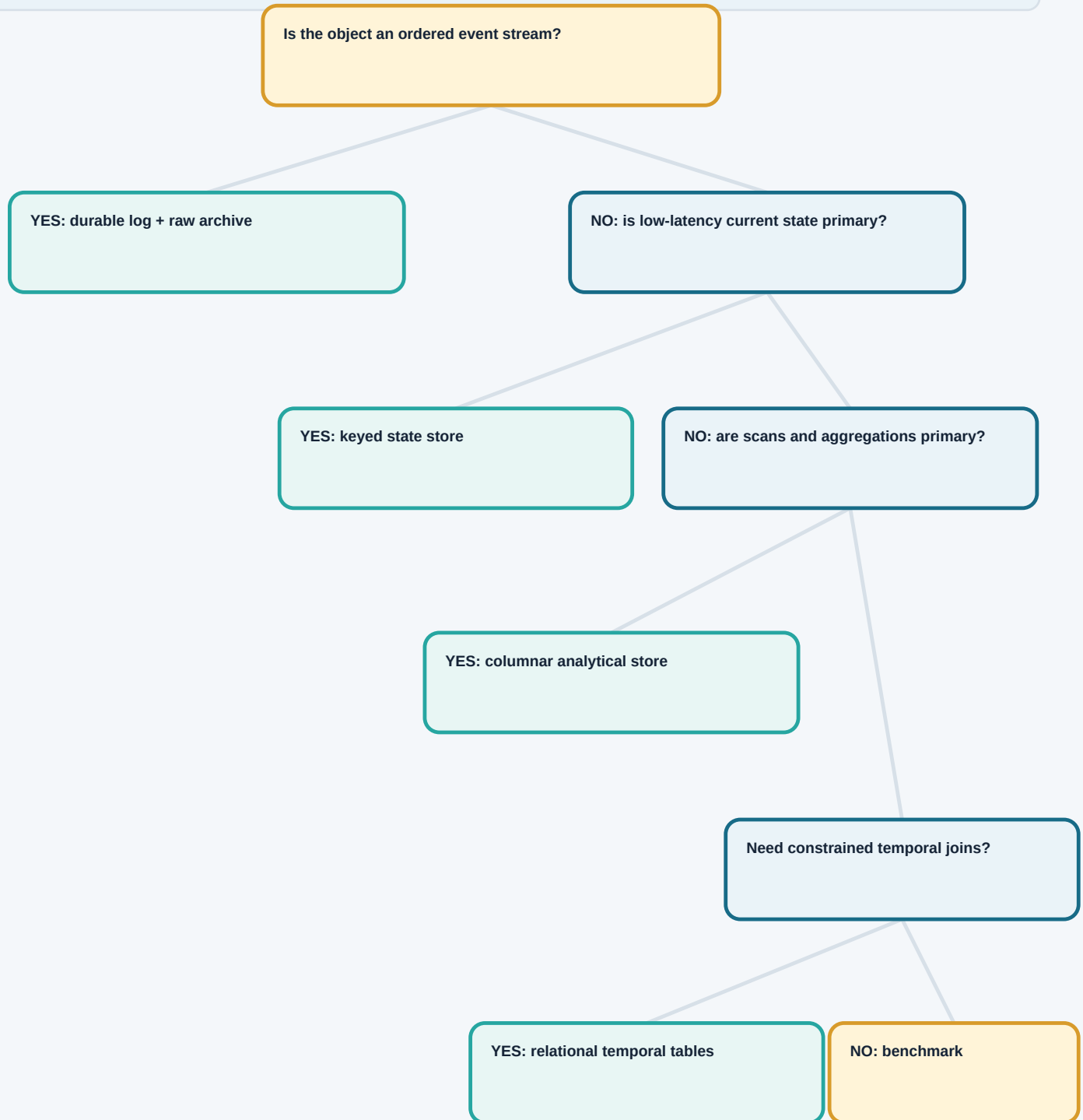
Choose a durable source representation and separate optimized serving views. Do not force every workload into one storage engine or one table shape.

Storage decision path

BEGINNER

QUESTION

Start from correctness, access pattern, and recovery. A technology name comes after the logical workload and failure contract.

**ARCHITECTURE RULE**

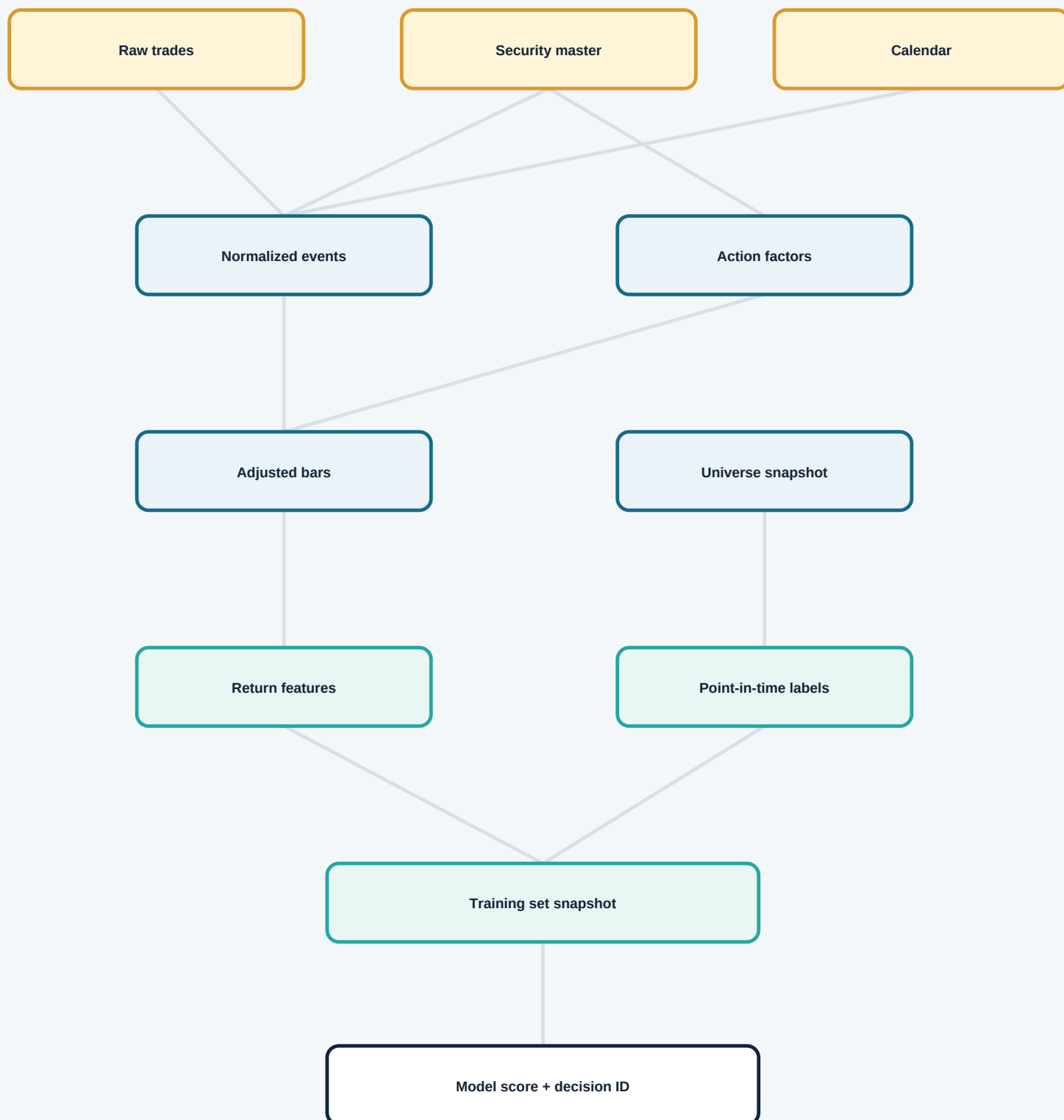
Most production platforms combine several roles and reconcile them through immutable IDs, manifests, and independent recomputation.

Lineage from trade to model decision

BEGINNER

PURPOSE

Every derived value should be traceable through exact datasets, transformations, parameters, and quality decisions. The graph also reveals blast radius when one source changes.



INCIDENT USE

A corrupted calendar identifies every affected bar, feature, training set, model, and decision without guesswork.

Freshness by source and session state

BEGINNER

INTERPRETATION

Cells show p95 observation age in seconds. A healthy threshold depends on the source and whether the market is open, auctioning, halted, or closed.

Trades	8.2s	0.4s	0.8s	0.3s	45.0s	999.0s
Top quotes	2.1s	0.1s	0.2s	0.1s	20.0s	999.0s
Depth	3.4s	0.2s	0.3s	0.2s	25.0s	999.0s
Reference	900.0s	900.0s	900.0s	900.0s	900.0s	900.0s
Actions	1.0h	1.0h	1.0h	1.0h	1.0h	1.0h
Fundamentals	>day	>day	>day	>day	>day	>day

DECISION READING

A fifteen-minute-old reference record may be healthy, while a fifteen-second-old top quote during continuous trading may be unusable. Closed-market age is expected and should not page an operator.

POLICY

Threshold by instrument liquidity, feed type, session state, and consumer tolerance. Publish age and reason code so downstream logic can abstain or fall back.

Failure modes and first diagnostic actions

BEGINNER

SYMPTOM	LIKELY CAUSE	CHECK	ACTION
Volume doubles after restart	Duplicate delivery	Event IDs and offsets	Replay idempotently
Negative spreads persist	Book sequence gap	Venue state and gaps	Rebuild from checkpoint
One-day -50% return	Unapplied split	Action effective date	Repair derived view
Backtest improves after reload	Mutable latest history	Snapshot and vintages	Bind immutable input
Intraday bars disagree	Boundary or eligibility	Source events and policy	Unify bar contract
Liquidity jumps 100x	Shares versus lots	Unit metadata	Version semantic schema
Feature rows multiply	Join cardinality	Keys and match counts	Aggregate before join
Live parity drifts	Batch/stream skew	Golden event replay	Share transforms
Feed healthy, venue absent	Coverage loss	Venue mix and counts	Degrade or switch

TRIAGE ORDER

Protect consumers, preserve raw evidence, isolate affected snapshots, verify identity/time/units, rebuild deterministically, reconcile, then republish under a new manifest.

EVENT LATENCY

Network delay= $t_{\text{arrival}} - t_{\text{event}}$; pipeline delay= $t_{\text{publish}} - t_{\text{arrival}}$. Keep clock source and synchronization quality beside each value.

QUOTE MEASURES

Mid=(bid+ask)/2; spread=ask-bid; imbalance=($Q_{\text{bid}} - Q_{\text{ask}}$)/($Q_{\text{bid}} + Q_{\text{ask}}$) over a declared depth set.

TRADE AGGREGATION

VWAP= $\sum \text{price}_i * \text{quantity}_i / \sum \text{quantity}_i$ for eligible trades. Arrival-time and event-time windows answer different questions.

OHLCV

Open=first, high=max, low=min, close=last, volume=sum over a precisely bounded eligible event set.

AS-OF MATCH

Choose the latest record with $\text{available_time} \leq \text{cutoff}$, valid identity, active effective interval, and age within tolerance.

FRESHNESS

age= $\text{decision_time} - \text{last_observed_time}$. Interpret age by source, instrument, session state, and expected update intensity.

NOTIONAL

notional= $\text{price} * \text{quantity} * \text{contract_multiplier}$, converted with a timestamped base/quote FX convention.

IDEMPOTENCE

For repeated event e , $f(f(S,e),e)=f(S,e)$. Stable identity and transactional publication make retry safe.

SNAPSHOT IDENTITY

Hash schema ID, ordered file checksums, source cutoffs, and transformation metadata to address a reproducible logical dataset.

Worked example: trade feed to training row

BEGINNER

1. CAPTURE EVENTS

Ingest venue trades with event, arrival, and normalize timestamps; source sequence and sale conditions remain attached to every raw payload.

2. RESOLVE IDENTITY

Map the venue symbol to an internal instrument using an effective-dated security master. Reject unresolved or overlapping mappings.

3. NORMALIZE UNITS

Convert size to shares and price to integer ticks while preserving original values, scale, currency, and contract version.

4. RECONCILE DELIVERY

Deduplicate by event identity, buffer bounded disorder, detect sequence gaps, and link corrections without erasing originals.

5. BUILD ONE-MINUTE BARS

Use left-closed intervals, regular eligible trades, exchange sessions, event-time ordering, and a stated late-event revision policy.

6. APPLY ACTIONS

Generate a split-adjusted view from point-in-time factors; retain raw bars and adjustment-factor lineage for every row.

7. CREATE FEATURES

At each decision cutoff, as-of join the last finalized bar and emit age, quality, session, snapshot, and transform IDs.

8. VALIDATE

Check keys, boundary fixtures, volume reconciliation, price bands, corporate-action continuity, missing sessions, and batch/stream parity.

9. PUBLISH AND REPLAY

Atomically publish an immutable manifest, bind the training set and model to its snapshot, and verify a raw-event-to-decision replay.

Final active recall and system index

BEGINNER

OBJECT MAP

Trades record executions; quotes record displayed intent; books encode depth and queue state; bars compress eligible paths; actions and reference data define interpretation.

TIME MAP

Event, arrival, process, publish, decision, route, and fill clocks support market order, causal eligibility, latency attribution, and replay.

IDENTITY MAP

Stable instrument IDs, venue IDs, effective mappings, event keys, sequences, and correction links prevent symbol reuse and duplicate state.

STRUCTURE MAP

Long, wide, nested, event, state, row, column, stream, and snapshot forms serve different operations under one semantic contract.

STORAGE MAP

Logs preserve order, row stores serve state, relational tables enforce temporal mappings, and columnar snapshots serve analytical scans.

QUALITY MAP

Missingness, staleness, outliers, crosses, sequence gaps, unit drift, coverage, survivorship, and mutable history need reason-coded controls.

REPRODUCIBILITY MAP

Schemas, code, parameters, source cutoffs, manifests, checksums, quality reports, and lineage bind every experiment to exact data.

PRODUCTION MAP

Observability, reconciliation, idempotent replay, fallback compatibility, rollback, ownership, and recovery tests keep data safe under failure.

FINAL RECALL

For any dataset, state its row grain, primary key, clocks, units, correction policy, point-in-time rule, snapshot identity, owner, and stop condition.